

Project 1 – Student Management System

Name: Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
Mini project.py > calculate_grade
1  # Student Management System
2  students = {}
3  def calculate_grade(per):
4      if per >= 90:
5          return "O"
6      elif per >= 80:
7          return "A+"
8      elif per >= 70:
9          return "A"
10     elif per >= 60:
11         return "B+"
12     elif per >= 50:
13         return "B"
14     else:
15         return "F"
16
17 def add_student():
18     try:
19         roll = int(input("Enter Roll No: "))
20
21         if roll in students:
22             print("Roll number already exists")
23             return
24
25         name = input("Enter Name: ")
26
27         marks = []
28         for i in range(5):
29             marks.append(float(input(f"Enter Subject {i+1} marks: ")))
30
```

```
31     percentage = sum(marks) / 5
32     grade = calculate_grade(percentage)
33
34     students[roll] = {
35         "name": name,
36         "marks": marks,
37         "percentage": percentage,
38         "grade": grade
39     }
40
41     print("Student Added Successfully")
42
43 except ValueError:
44     print("Invalid Input")
45
46
47 def view_all():
48     for roll, data in students.items():
49         print("\nRoll:", roll)
50         print("Name:", data["name"])
51         print("Percentage:", data["percentage"])
52         print("Grade:", data["grade"])
53
54
55 def search_student():
56     roll = int(input("Enter Roll No: "))
57
58     if roll in students:
59         print(students[roll])
60     else:
61         print("Student Not Found")
62
63
64 def update_student():
65     roll = int(input("Enter Roll No: "))
66
```

```

67     if roll in students:
68         name = input("Enter New Name: ")
69         students[roll]["name"] = name
70         print("Updated Successfully")
71     else:
72         print("Not Found")
73
74
75     def delete_student():
76         roll = int(input("Enter Roll No: "))
77
78         if roll in students:
79             del students[roll]
80             print("Deleted Successfully")
81         else:
82             print("Not Found")
83
84
85     while True:
86
87         print("""
88         ===== STUDENT MANAGEMENT SYSTEM =====
89         1.Add Student
90         2.View All
91         3.Search
92         4.Update
93         5.Delete
94         6.Exit
95         """)
96
97         choice = int(input("Enter Choice: "))
98
99         if choice == 1:
100             add_student()
101
102         elif choice == 2:

```

```

102         elif choice == 2:
103             view_all()
104
105         elif choice == 3:
106             search_student()
107
108         elif choice == 4:
109             update_student()
110
111         elif choice == 5:
112             delete_student()
113
114         elif choice == 6:
115             break
116
117         else:
118             print("Wrong Choice")

```

Output:

```
===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 1
Enter Roll No: 14
Enter Name: shruti
Enter Subject 1 marks: 80
Enter Subject 2 marks: 77
Enter Subject 3 marks: 67
Enter Subject 4 marks: 87
Enter Subject 5 marks: 91
Student Added Successfully

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 2

Roll: 14
Name: shruti
Percentage: 80.4
Grade: A+

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 3
Enter Roll No: 25
Student Not Found
```

```
===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 4
Enter Roll No: 14
Enter New Name: shruti
Updated Successfully

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 5
Enter Roll No: 16
Not Found

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 6
PS C:\Users\tanve\Desktop\ITR> 
```

Project 2 – library Management System

Name: Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
Mini project.py > ...
1  # Library Management System
2
3  books = {}
4
5  def add_book():
6      isbn = input("Enter ISBN: ")
7
8      if isbn in books:
9          print("Book already exists")
10         return
11
12     title = input("Enter Book Title: ")
13     author = input("Enter Author: ")
14
15     books[isbn] = {
16         "title": title,
17         "author": author,
18         "available": True,
19         "borrower": None
20     }
21
22     print("Book Added Successfully")
23
24
25  def view_books():
26      for isbn, book in books.items():
27          print("\nISBN:", isbn)
28          print("Title:", book["title"])
29          print("Author:", book["author"])
30          print("Status:",
31                "Available" if book["available"] else "Issued")
32
33
34  def search_book():
35      key = input("Enter title/author: ").lower()
36
37      for book in books.values():
38          if key in book["title"].lower() or key in book["author"].lower():
39              print(book)
40
41
42  def issue_book():
43      isbn = input("Enter ISBN: ")
44
```

```

Mini project.py > ...
42 def issue_book():
43     ...
44
45     if isbn in books:
46
47         if books[isbn]["available"]:
48             name = input("Enter Borrower Name: ")
49
50             books[isbn]["available"] = False
51             books[isbn]["borrower"] = name
52
53             print("Book Issued Successfully")
54
55         else:
56             print("Book already issued")
57
58     else:
59         print("Book not found")
60
61
62 def return_book():
63     isbn = input("Enter ISBN: ")
64
65     if isbn in books:
66
67         books[isbn]["available"] = True
68         books[isbn]["borrower"] = None
69
70         print("Book Returned")
71
72     else:
73         print("Book not found")
74
75
76 while True:
77
78     print("""
79     ===== LIBRARY MANAGEMENT SYSTEM =====
80
81     1.Add Book
82     2.Issue Book
83     3.Return Book
84     4.Search Book
85     5.View Books
86     6.Exit

```

```

78     print(
79     ===== LIBRARY MANAGEMENT SYSTEM =====
80
81     1.Add Book
82     2.Issue Book
83     3.Return Book
84     4.Search Book
85     5.View Books
86     6.Exit
87
88     """)
89
90     choice = int(input("Enter Choice: "))
91
92     if choice == 1:
93         add_book()
94
95     elif choice == 2:
96         issue_book()
97
98     elif choice == 3:
99         return_book()
100
101     elif choice == 4:
102         search_book()
103
104     elif choice == 5:
105         view_books()
106
107     elif choice == 6:
108         break
109
110     else:
111         print("Invalid Choice")

```

Output:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Python\pythoncore-  
'--' 'c:\Users\tanve\Desktop\ITR\Mini project.py'  
  
===== LIBRARY MANAGEMENT SYSTEM =====  
  
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit  
  
Enter Choice: 101  
Invalid Choice  
  
===== LIBRARY MANAGEMENT SYSTEM =====  
  
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit  
  
Enter Choice: 1  
Enter ISBN: 101  
Enter Book Title: python  
Enter Author: Martin c. Brown  
Book Added Successfully  
  
===== LIBRARY MANAGEMENT SYSTEM =====  
  
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit  
  
Enter Choice: 1  
Enter ISBN: 102  
Enter Book Title: Beloved  
Enter Author: Toni Morrisson  
Book Added Successfully
```

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Python
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 1  
Enter ISBN: 104  
Enter Book Title: Chaitali  
Enter Author: R.N.Tagore  
Book Added Successfully
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 2  
Enter ISBN: 102  
Enter Borrower Name: Shruti  
Book Issued Successfully
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 2  
Enter ISBN: 104  
Enter Borrower Name: sairaj  
Book Issued Successfully
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 3  
Enter ISBN: 101  
Book Returned
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 4  
Enter title/author: chaitali  
{'title': 'Chaitali', 'author': 'R.N.Tagore', 'available': False, 'borrower': 'sairaj'}
```

```
===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1.Add Book  
2.Issue Book  
3.Return Book  
4.Search Book  
5.View Books  
6.Exit
```

```
Enter Choice: 5
```

```
ISBN: 101  
Title: python  
Author: Martin c. Brown  
Status: Available
```

```
ISBN: 102  
Title: Beloved  
Author: Toni Morrisson
```

```
Enter Choice: 5

ISBN: 101
Title: python
Author: Martin c. Brown
Status: Available

ISBN: 102
Title: Beloved
Author: Toni Morrisson
Status: Issued

ISBN: 104
Title: Chaitali
Author: R.N.Tagore
Status: Issued

===== LIBRARY MANAGEMENT SYSTEM =====

1.Add Book
2.Issue Book
3.Return Book
4.Search Book
5.View Books
6.Exit

Enter Choice: 6
PS C:\Users\tanve\Desktop\ITR> 
```

Project 3 – Personal Expense Tracker

Name: Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
project 3.py > ...
1  # Personal Expense Tracker
2
3  expenses = []
4
5  budget = float(input("Enter Monthly Budget: "))
6
7
8  def add_expense():
9
10     try:
11         desc = input("Enter Description: ")
12         category = input("Enter Category: ")
13         amount = float(input("Enter Amount: "))
14         date = input("Enter Date (DD-MM-YYYY): ")
15
16         if amount <= 0:
17             print("Invalid Amount")
18             return
19
20         expense = {
21             "description": desc,
22             "category": category,
23             "amount": amount,
24             "date": date
25         }
26
27         expenses.append(expense)
28
29         print("Expense Added Successfully")
30
```

```

31     except ValueError:
32         print("Enter valid amount")
33
34
35 def view_expenses():
36
37     if len(expenses) == 0:
38         print("No Expenses")
39
40     else:
41         for i, e in enumerate(expenses,1):
42
43             print("\nExpense No:", i)
44             print("Description:", e["description"])
45             print("Category:", e["category"])
46             print("Amount:", e["amount"])
47             print("Date:", e["date"])
48
49
50
51 def category_summary():
52
53     summary = {}
54
55     for e in expenses:
56
57         cat = e["category"]
58
59         if cat in summary:
60             summary[cat] += e["amount"]
61
62         else:
63             summary[cat] = e["amount"]
64
65
66     print("\nCategory Summary")
67
68     for c, total in summary.items():
69         print(c, ":", total)
70

```

```

project 3.py > ...
73 def budget_report():
74
75     total = 0
76
77     for e in expenses:
78         total += e["amount"]
79
80
81     remaining = budget - total
82
83     used = (total / budget) * 100
84
85
86     print("""
87 ===== BUDGET REPORT =====
88 """)
89
90     print("Total Spent:", total)
91     print("Budget:", budget)
92     print("Remaining:", remaining)
93     print("Used:", used, "%")
94
95     if used >= 100:
96         print("WARNING: Budget Exceeded")
97
98     elif used >= 80:
99         print("WARNING: 80% Budget Used")
100
101
102
103 while True:
104
105     print("""
106 ===== EXPENSE TRACKER =====
107
108 1.Add Expense
109 2.View Expenses
110 3.Category Summary
111 4.Budget Report
112 5.Exit
113
114 """)
115
116     choice = int(input("Enter Choice: "))

```

```
114 """  
Click to add a breakpoint  
116 choice = int(input("Enter Choice: "))  
117  
118  
119 if choice == 1:  
120     add_expense()  
121  
122  
123 elif choice == 2:  
124     view_expenses()  
125  
126  
127 elif choice == 3:  
128     category_summary()  
129  
130  
131 elif choice == 4:  
132     budget_report()  
133  
134  
135 elif choice == 5:  
136     break  
137  
138  
139 else:  
140     print("Invalid Choice")
```

Output:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.9.0.0_qazws8x21c56tc6wsc55k358win7cp301.exe' -- 'c:\Users\tanve\Desktop\ITR\project 3.py'
Enter Monthly Budget: 2500

===== EXPENSE TRACKER =====

1.Add Expense
2.View Expenses
3.Category Summary
4.Budget Report
5.Exit

Enter Choice: 1
Enter Description: Burger
Enter Category: Food
Enter Amount: 199
Enter Date (DD-MM-YYYY): 22
Expense Added Successfully

===== EXPENSE TRACKER =====

1.Add Expense
2.View Expenses
3.Category Summary
4.Budget Report
5.Exit

Enter Choice: 1
Enter Description: Bus ticket
Enter Category: Travel
Enter Amount: 50
Enter Date (DD-MM-YYYY): 22-06-2026
Expense Added Successfully

===== EXPENSE TRACKER =====

1.Add Expense
2.View Expenses
3.Category Summary
4.Budget Report
5.Exit
```

Enter Choice: 2

Expense No: 1

Description: Burger

Category: Food

Amount: 199.0

Date: 22

Expense No: 2

Description: Bus ticket

Category: Travel

Amount: 50.0

Date: 22-06-2026

===== EXPENSE TRACKER =====

1.Add Expense

2.View Expenses

3.Category Summary

4.Budget Report

5.Exit

Enter Choice: 3

Category Summary

Food : 199.0

Travel : 50.0

===== EXPENSE TRACKER =====

1.Add Expense

2.View Expenses

3.Category Summary

4.Budget Report

5.Exit

Enter Choice: 4

===== BUDGET REPORT =====

Total Spent: 249.0

Budget: 2500.0

Remaining: 2251.0

Used: 9.959999999999999 %

Enter Choice: 4

===== BUDGET REPORT =====

Total Spent: 249.0

Budget: 2500.0

Remaining: 2251.0

Used: 9.959999999999999 %

===== EXPENSE TRACKER =====

- 1.Add Expense
- 2.View Expenses
- 3.Category Summary
- 4.Budget Report
- 5.Exit

Enter Choice: 5

PS C:\Users\tanve\Desktop\ITR> █

Project 4 – Quiz and Examination System

Name: Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
1  # Quiz and Examination System
2
3  questions = [
4      ("Which data type is immutable?", "List", "Dictionary", "Tuple", "Set", "C"),
5      ("Which keyword is used for function?", "define", "func", "def", "function", "C"),
6      ("Which loop repeats until condition becomes false?", "for", "while", "switch", "case", "B"),
7      ("Python file extension?", ".java", ".py", ".cpp", ".html", "B"),
8      ("Which data structure stores key-value pairs?", "List", "Tuple", "Dictionary", "Set", "C"),
9      ("Which operator is used for exponent?", "+", "*", "**", "/", "C"),
10     ("Input function in Python?", "print()", "scan()", "input()", "read()", "C"),
11     ("Which statement is used for decision making?", "if", "for", "while", "loop", "A"),
12     ("Which collection stores unique values?", "List", "Tuple", "Set", "String", "C"),
13     ("Which keyword exits a loop?", "skip", "break", "stop", "exit", "B")
14 ]
15
16 def calculate_grade(percent):
17     if percent >= 90:
18         return "O"
19     elif percent >= 80:
20         return "A+"
21     elif percent >= 70:
22         return "A"
23     elif percent >= 60:
24         return "B+"
25     elif percent >= 50:
26         return "B"
27     else:
28         return "F"
29
30 name = input("Enter Student Name: ")
31 roll = input("Enter Roll Number: ")
32
```

```

33 score = 0
34 wrong_answers = []
35
36 for i, q in enumerate(questions, 1):
37     print(f"\nQ{i}. {q[0]}")
38     print("A.", q[1])
39     print("B.", q[2])
40     print("C.", q[3])
41     print("D.", q[4])
42
43     answer = input("Your Answer (A/B/C/D): ").upper()
44
45     while answer not in ["A", "B", "C", "D"]:
46         answer = input("Enter only A/B/C/D: ").upper()
47
48     if answer == q[5]:
49         print("✓ Correct")
50         score += 1
51     else:
52         print("X Incorrect")
53         wrong_answers.append((q[0], q[5]))
54
55
56 total = len(questions)
57 percent = (score / total) * 100
58 grade = calculate_grade(percent)
59
60 print("\n===== RESULT REPORT =====")
61 print("Name :", name)
62 print("Roll :", roll)
63 print("Score :", score, "/", total)
64 print("Percentage :", round(percent, 2), "%")
65 print("Grade :", grade)
66
67 if percent >= 50:
68     print("Result : PASS")
69 else:
70     print("Result : FAIL")
71
72 if wrong_answers:
73     print("\nWrong Answers:")
74     for q, ans in wrong_answers:
75         print("Question:", q)
76         print("Correct Answer:", ans)

```

Output:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\Desktop\ITR'
ndled\libs\debugpy\launcher' '51330' '--' 'c:\Users\tanve\Desktop\I
Enter Student Name: shruti
Enter Roll Number: 14

Q1. Which data type is immutable?
A. List
B. Dictionary
C. Tuple
D. Set
Your Answer (A/B/C/D): c
✓ Correct

Q2. Which keyword is used for function?
A. define
B. func
C. def
D. function
Your Answer (A/B/C/D): c
✓ Correct

Q3. Which loop repeats until condition becomes false?
A. for
B. while
C. switch
D. case
Your Answer (A/B/C/D): A
X Incorrect

Q4. Python file extension?
A. .java
B. .py
C. .cpp
D. .html
Your Answer (A/B/C/D): B
✓ Correct

Q5. Which data structure stores key-value pairs?
A. List
B. Tuple
C. Dictionary
D. Set
Your Answer (A/B/C/D): C
✓ Correct
```

Q6. Which operator is used for exponent?

- A. +
- B. *
- C. **
- D. /

Your Answer (A/B/C/D): B

X Incorrect

Q7. Input function in Python?

- A. print()
- B. scan()
- C. input()
- D. read()

Your Answer (A/B/C/D): C

✓ Correct

Q8. Which statement is used for decision making?

- A. if
- B. for
- C. while
- D. loop

Your Answer (A/B/C/D): A

✓ Correct

Q9. Which collection stores unique values?

- A. List
- B. Tuple
- C. Set
- D. String

Your Answer (A/B/C/D): C

✓ Correct

Q10. Which keyword exits a loop?

- A. skip
- B. break
- C. stop
- D. exit

Your Answer (A/B/C/D): B

✓ Correct

===== RESULT REPORT =====

Name : shruti

Roll : 14

Score : 8 / 10

Percentage : 80.0 %

Grade : A+

Result : PASS

```
===== RESULT REPORT =====
```

```
Name : shruti
```

```
Roll : 14
```

```
Score : 8 / 10
```

```
Percentage : 80.0 %
```

```
Grade : A+
```

```
Result : PASS
```

```
Wrong Answers:
```

```
Question: Which loop repeats until condition becomes false?
```

```
Correct Answer: B
```

```
Question: Which operator is used for exponent?
```

```
Correct Answer: C
```

```
PS C:\Users\tanve\Desktop\ITR> |
```

Project 5 – Inventory Management System

Name: Shruti Dnyaneshwar Darwatkar

Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
1  # Inventory Management System
2
3  inventory = {}
4  transactions = []
5
6
7  def add_product():
8
9      pid = input("Enter Product ID: ")
10
11      if pid in inventory:
12          print("Product already exists")
13          return
14
15      name = input("Enter Product Name: ")
16      category = input("Enter Category: ")
17      price = float(input("Enter Price: "))
18      quantity = int(input("Enter Quantity: "))
19      reorder = int(input("Enter Reorder Level: "))
20
21
22      inventory[pid] = {
23          "name": name,
24          "category": category,
25          "price": price,
26          "quantity": quantity,
27          "reorder": reorder
28      }
29
30      print("Product Added Successfully")
31
32
33
```

```

34 def stock_in():
35
36     pid = input("Enter Product ID: ")
37
38     if pid in inventory:
39
40         qty = int(input("Enter Quantity: "))
41
42         inventory[pid]["quantity"] += qty
43
44         transactions.append(
45             f"IN : {pid} +{qty}"
46         )
47
48         print("Stock Added")
49
50     else:
51         print("Product Not Found")
52
53
54
55 def stock_out():
56
57     pid = input("Enter Product ID: ")
58
59     if pid in inventory:
60
61         qty = int(input("Enter Quantity: "))
62
63
64         if inventory[pid]["quantity"] >= qty:
65
66             inventory[pid]["quantity"] -= qty
67
68             transactions.append(
69                 f"OUT : {pid} -{qty}"
70             )
71
72             print("Stock Removed")
73
74         else:
75             print("Not enough stock")
76

```

```

76
77     else:
78         print("Product Not Found")
79
80
81
82 def view_inventory():
83
84     for pid, p in inventory.items():
85
86         value = p["price"] * p["quantity"]
87
88         print("\nProduct ID:", pid)
89         print("Name:", p["name"])
90         print("Category:", p["category"])
91         print("Price:", p["price"])
92         print("Quantity:", p["quantity"])
93         print("Total Value:", value)
94
95
96
97 def low_stock_alert():
98
99     print("\nLow Stock Products")
100
101     for pid,p in inventory.items():
102
103         if p["quantity"] <= p["reorder"]:
104             print(p["name"], "- Quantity:", p["quantity"])
105
106
107
108 def generate_report():
109
110     total_value = 0
111     categories = set()
112
113
114     for p in inventory.values():
115
116         total_value += p["price"] * p["quantity"]
117         categories.add(p["category"])
118

```

```

120 > print("""===== INVENTORY REPORT ===== ...
122
123     print("Total Products:", len(inventory))
124     print("Total Stock Value:", total_value)
125     print("Categories:", categories)
126
127
128     print("\nTransaction History")
129
130     for t in transactions:
131         print(t)
132
133
134
135 while True:
136
137     print("""
138     ===== INVENTORY MANAGEMENT SYSTEM =====
139
140     1.Add Product
141     2.Stock In
142     3.Stock Out
143     4.View Inventory
144     5.Low Stock Alert
145     6.Report
146     7.Exit
147
148     """)
149
150     choice = int(input("Enter Choice: "))
151
152
153     if choice == 1:
154         add_product()
155
156
157     elif choice == 2:
158         stock_in()
159
160
161     elif choice == 3:
162         stock_out()
163

```

```
150     choice = int(input("Enter Choice: "))
151
152
153     if choice == 1:
154         add_product()
155
156
157     elif choice == 2:
158         stock_in()
159
160
161     elif choice == 3:
162         stock_out()
163
164
165     elif choice == 4:
166         view_inventory()
167
168
169     elif choice == 5:
170         low_stock_alert()
171
172
173     elif choice == 6:
174         generate_report()
175
176
177     elif choice == 7:
178         break
179
180
181     else:
182         print("Invalid Choice")
183
```

Output:

```
PS C:\Users\tanve\Desktop\ITR> c;; cd 'c:\Users\tanve\Desktop\I
ndled\libs\debugpy\launcher' '59752' '--' 'c:\Users\tanve\Desktop

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice:

      1
Enter Product ID: s123
Enter Product Name: Mobile
Enter Category: Electronic
Enter Price: 30000
Enter Quantity: 1
Enter Reorder Level: 1
Product Added Successfully

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 4

Product ID: s123
Name: Mobile
Category: Electronic
Price: 30000.0
Quantity: 1
Total Value: 30000.0
```

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report
- 7.Exit

Enter Choice: 4

Product ID: s123
Name: Mobile
Category: Electronic
Price: 30000.0
Quantity: 1
Total Value: 30000.0

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report
- 7.Exit

Enter Choice: 2
Enter Product ID: s123
Enter Quantity: 2
Stock Added

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report

```
6.Report
7.Exit

Enter Choice: 3
Enter Product ID: s123
Enter Quantity: 1
Stock Removed

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 5

Low Stock Products

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 6
===== INVENTORY REPORT =====

Total Products: 1
Total Stock Value: 60000.0
Categories: {'Electronic'}

Transaction History
IN : s123 +2
```

```
Enter Choice: 6
===== INVENTORY REPORT =====

Total Products: 1
Total Stock Value: 60000.0
Categories: {'Electronic'}

Transaction History
IN : s123 +2
OUT : s123 -1

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 7
PS C:\Users\tanve\Desktop\ITR> |
```